

COMMUNICATION OF EMERGENCY PUBLIC WARNINGS

**A Social Science Perspective and
State-of-the-Art Assessment**

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ABSTRACT

More than 200 studies of warning systems and warning response were reviewed for this social science perspective and state-of-the-art assessment of communication of emergency public warnings. The major findings are as follows.

First, variations in the nature and content of warnings have a large impact on whether or not the public heeds the warning. Relevant factors include the warning source; warning channel; the consistency, credibility, accuracy, and understandability of the message; and the warning frequency.

Second, characteristics of the population receiving the warning affect warning response. These include social characteristics such as gender, ethnicity and age, social setting characteristics such as stage of life or family context, psychological characteristics such as fatalism or risk perception, and knowledge characteristics such as experience or training.

Third, many current myths about public response to emergency warning are at odds with knowledge derived from field investigations. Some of these myths include the "keep it simple" notion, the "cry wolf" syndrome, public panic and hysteria, and those concerning public willingness to respond to warnings.

Finally, different methods of warning the public are not equally effective at providing an alert and notification in different physical and social settings. Most systems can provide a warning given three or more hours of available warning time. Special systems such as tone-alert radios are needed to provide rapid warning.

Either an individual or a group can have warning decision authority. If this authority rests with an individual, a back-up decision structure should be specified in case that individual is unavailable, and if decision authority rests with a group, the membership and convening mechanism should be established as well as backup procedures should the group be unable to convene. Who decides may be determined by legal mandate. In some states, only the governor can legally issue a warning; in others, the person in authority may be a county sheriff or a local mayor. In any case, planners should ensure that a prompt decision can be made if the situation calls for a rapid warning.

3.4.3 Decision-Making Processes

It is also desirable to specify in plans how the warning decision will be made in the emergency situation. This involves establishing the broad criteria on which to make a decision and indicating how those criteria should be used. Rigid decision-making frameworks should be avoided. Human judgment is still an important and necessary part of decision making even with today's advanced technology.

Analytical models and decision criteria are helpful to making good decisions, but these tools cannot make the decision. For example, one warning decision system we reviewed involved a complex model in which data were entered and the system made a yes or no warning decision. But as the final step of the process the decision maker could override the model and go ahead with the warning anyway. Decision models may be of greater use in deciding when and where to warn than in actually deciding whether to warn. The exception would be for extremely fast-moving events in which a warning must be automatically triggered to provide sufficient time for the public to take protective actions.

3.5 WRITING THE WARNING MESSAGE

One of the clearest and most consistent conclusions of social science research is that the warning message itself is one of the most important factors in determining the effectiveness of a warning system. In large part, it is the content and style of the actual warning message that shapes the extent to which an endangered public engages in protective actions.

In the following section, we review the elements of both message style and content that should be considered in writing a public warning. Before proceeding, however, let us correct the notion that public warning messages must be short or else the endangered public will become confused or lose interest in the subject. The public does have a short attention span. But major emergencies like tsunamis, dam failures, and nuclear power plant accidents are unique in terms of how willing a public is to listen to information. Emergency warnings of impending catastrophes convert an information-adverse public (you have only 30 seconds to convince me to buy your product) into a public that is information hungry (why are we at risk, do you really mean me, how long do I have, what is it you think I should do, and so on). Warning messages that "keep it short" are inappropriate in public emergencies because short messages set a diverse at-risk public on an information scavenger hunt to fill the information voids left by the short message. Such brief messages can be dangerous since they can lead people to consult friends, neighbors, relatives, superstitions, biases, and a raft of other "information providers" to fill the void. These other sources may provide inaccurate information (it never floods here, lightning never strikes in the same place twice and we had ours last year, all nuclear power accidents release radiation as happened at Chernobyl) and create rumors. Subsequently,

poor public response decisions or lack of protective actions can result. The sections which follow address the style and content of public emergency messages appropriate for inclusion in plans for warnings.

3.5.1 The Warning Content

Five topics are important to consider in assembling the content of a public warning message. These topics are hazard or risk, guidance, location, time, and source (Fig. 3.2).

3.5.1.1 Hazard

A warning message must provide the public with information about the impending hazard by describing the event that may occur and how it poses a danger to people. It is insufficient, for example, for a warning to simply state that a dam may break. This warning must also describe the height and speed of impact of the floodwaters that will ensue, and the size and location of the areas that could be affected. A warning for a nuclear power plant accident might indicate that the radiation will filter into the air like a cloud and then travel with the wind while becoming less and less concentrated.

These examples are not meant as prototype descriptions for dam failure and nuclear power plant radiation releases. They simply illustrate that warnings should be specific about the character of the hazards involved. A warning could describe "a wall of water 20 feet high moving at 40 miles per hour," "an explosion hotter than the inside of the sun covering half of the county," "or a seismic shake severe enough to bring down half the unreinforced brick buildings in the city." If a hazard is well described, people are better able to understand the logic of protective actions, (e.g., close the windows in the house because the risk is in the air; get out of brick buildings because they may fall down).

Thus, hazards should be described with sufficient detail so that all members of the public understand the character of the disaster agent from which they are to protect themselves. Informing the public about the physical characteristics of the hazard will reduce the number of people in an endangered public who misperceive the hazard and then make poor response decisions because of those misperceptions. The hazard aspect of warning message content provides the public with a rationale for subsequent behavior.

3.5.1.2 Guidance

Public warning messages must also include guidance about what people should do to maximize their safety in the face of impending disaster. It cannot be assumed that members of the public will know what constitutes an appropriate protective action. The protective action must be described. This point may seem obvious, but it is not. For example, warnings must do more than tell people that they should "get to high ground." High ground for some may be the low ground for others. High ground should be defined—for example, "ground higher than the top of City Hall," or specify areas to which people should evacuate.

MESSAGE STYLE	MESSAGE CONTENT				
	HAZARD	LOCATION	GUIDANCE	TIME	SOURCES
SPECIFICITY					
CONSISTENCY					
ACCURACY					
CERTAINTY					
CLARITY					

Fig. 3.2. The style and content of a warning message.

3.5.1.3 Location

A warning message must also describe the location of risk because of the impending hazard. The hazard factor first described and this location factor are closely linked. Detailing the location of risk is best done in ways readily understood by the public. For example, a flash flood warning could say: "The area of town that will flood will be between Second and Fifth Streets, from Elm Avenue to Magnolia Boulevard." If there is reason to be concerned that people who are safe could think that they are unsafe, then the warning should address them—for example, "People who live in other parts of the city will not experience flooding"—but the warning should then explain why they are safe. This is usually necessary because a wider audience than those at risk will hear the warning message.

3.5.1.4 Time

Public warnings must also address the "when" aspect of response. The public at risk needs information about how much time is available for them to engage in protective actions before impact, or how much time there is before they should initiate protective actions. For example, "The tsunamis would not strike before 10 p.m. this evening, and you should be on the northern side of U.S. Highway 72 by 9:45 p.m. to be on the safe side."

3.5.1.5 Source

The final important dimension of warning content is the source of the warning. The source of the warning should be identified in the message. Warnings are most believable if they come from a mixed set of persons. For example, "The mayor and the head of civil defense have just conferred with scientists from our local university and the National Weather Service as well as with the head of our local Red Cross chapter, and we now wish to warn you that. . . ."

3.5.2 The Warning Style

The five aspects of warning content can be cross-classified against the varied stylistic aspects of a warning message (Fig. 3.2). The stylistic aspects are specificity, consistency, accuracy, certainty, and clarity. A warning message could readily be evaluated by viewing the specificity of the message regarding location, guidance, hazard and time; the consistency of the message regarding these same content factors; and so on. The sections which follow describe the quality of the five stylistic aspects of the most effective public warnings.

3.5.2.1 Specificity

A good warning message is specific about the area at risk, what people should do, the character of the hazard, how much time people have to engage in protective actions, and the source of the message. There are many occasions when specificity on all these items cannot be high. Something may be unknown or known imprecisely. On these occasions, the warning message itself need not be nonspecific. For example, "We do not know nor can it be known which buildings in the city will be safe and which will not be

safe when the earthquake strikes, but we do know that most people will be safer if they go home now."

3.5.2.2 Consistency

A warning message must also be consistent, both within itself as well as across different messages. Inconsistencies exist within a message for a variety of reasons and in different ways. For example, it is inconsistent to tell a public to evacuate but that their children will be kept in neighborhood schools. In most emergencies there are numerous inconsistencies across different warnings as more is learned about the impending event and updates are issued. For example, inconsistencies can appear as new information reveals that the hazard has decreased or increased, the number of people at risk has become larger or smaller, and so on. Consistency can be rendered across messages in circumstances such as these by simply referencing and repeating what was last said, what has changed, and why.

3.5.2.3 Certainty

A message should be stated with certainty even in circumstances in which there is ambiguity associated with the hazard's impact. For example, "There is no way for us to know if there really is a bomb in the skyscraper, or that it will actually go off at 3 p.m. if there is, but we have decided to recommend that the building be evacuated now, and that we will act as if the bomb threat is a real one." Certainty in warning messages extend beyond message content to include the tone with which it is delivered to the public. The warning should be spoken by the person delivering it as if he or she believes or is certain about what is being said.

3.5.2.4 Clarity

Warnings must be worded in simple language that can be understood. For example, "a possible transient excursion of the reactor resulting in a sudden relocation of the core materials outside the containment vessel" might better be stated as "some radiation may escape from a hole in the nuclear reactor."

3.5.2.5 Accuracy

The last important stylistic attribute is accuracy. A warning message must contain timely, accurate, and complete information. If people learn or suspect that they are not receiving the whole truth, they may well not believe the message, or they may consider its sources to be noncredible. Accuracy is enhanced simply by being fully open and honest with the public regarding a hazard. In addition, accuracy is important in parts of the warning that may be viewed by officials as being trivial. For example, calling Broad Street "Board" Street by mistake may send a signal to the public that other essential information is also incorrect, even though they can correct the error on the basis of personal knowledge.